Silvestre Ronin, silver3@rpi.edu, 662043925
Kurbiel Chris, kurbic@rpi.edu, 662041666
Homework 2
Professor Radev
COGNITIVE MODELING - COGS 4210 - 01
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1 Problem 1 [FINISHED]

All statements that are false: 1, 3, 4, 6, 7.

2 Problem 2 [FINISHED]

Let X be a random variable denoting the oil market value in relation to its original market value. Then, we have that

$$\mathbb{E}[X] = (0.8)(1.01) + (0.2)(0.9) = 0.988$$

So, we have that $\mathbb{E}[X] = 0.988 < 1$, which implies that the original market value has gone down. Hence, we would not invest in this market.

Assuming that the increase/drop in prices is fixed, we will now calculate the minimal probability of prices going up, $\mathbf{p}$, in order for us to invest rationally in this market, according to expectation. It holds that we would invest in the market for $\mathbb{E}[X] > 1$ as it implies that the original market value has gone up. Thus, we have that

$$\begin{aligned}
\mathbf{p}(1.01) + (1 - \mathbf{p})(0.9) &> 1 \\
\mathbf{p}(1.01) + 0.9 - \mathbf{p}(0.9) &> 1 \\
(1.01 - 0.9)\mathbf{p} &> 1 - 0.9 \\
\mathbf{p} &> \frac{.10}{.11} \\
\mathbf{p} &> \frac{10}{11}
\end{aligned}$$

Thus, we conclude that for $\mathbf{p} > \frac{10}{11}$, we would invest in this market.

The limitation of expectations when making single-shot, real-life decisions is that expectations is about average outcome over many repeated trials, but in a single-shot decision the distribution (especially the downside risk) matters, and not just the mean.

3 Problem 3 [FINISHED]

1. Show the following identity for the variance of a random variable X:

$$\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

We will show that for the variance of a random variable X, the property $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ holds.

We have that $\text{Var}[X] = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$. Now, we denote $\mu = \mathbb{E}[X]$. Then, we simplify as follows:

$$\begin{aligned}
\text{Var}[X] &= \mathbb{E}\left[(X-\mu)^2\right] \\
&= \mathbb{E}\left[X^2 - 2X\mu + \mu^2\right] \\
&= \mathbb{E}[X^2] - \mathbb{E}[2X\mu] + \mathbb{E}[\mu^2] \\
&= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mathbb{E}[\mu^2] \\
&= \mathbb{E}[X^2] - 2\mu^2 + \mathbb{E}[\mu^2] \\
&= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 \\
&= \mathbb{E}[X^2] - \mu^2 \\
&= \mathbb{E}[X^2] - \mathbb{E}[X]^2.
\end{aligned}$$

Thus, we have shown that $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

2. Show the following property for the variance of a random variable X and a scalar α:

$$\text{Var}[\alpha X + \beta] = \alpha^2 \text{Var}[X]$$

We will show that for the variance of a random variable X and a scalar α, the property $\text{Var}[\alpha X + \beta] = \alpha^2 \text{Var}[X]$ holds.

From part 1), we have that $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. Now, we denote $\mu = \mathbb{E}[X]$. Then, we simplify as follows:

$$
\begin{aligned}
Var[\alpha X + \beta] &= \mathbb{E}[(\alpha X + \beta)^2] - \mathbb{E}[\alpha X + \beta]^2 \\
&= \mathbb{E}[\alpha^2 X^2 + 2\alpha X\beta + \beta^2] - (\mathbb{E}[\alpha X + \beta])^2 \\
&= \mathbb{E}[\alpha^2 X^2] + \mathbb{E}[2\alpha X\beta] + \mathbb{E}[\beta^2] - (\mathbb{E}[\alpha X + \beta])^2 \\
&= \alpha^2\mathbb{E}[X^2] + 2\alpha\beta\mu + \beta^2 - (\alpha\mathbb{E}[X] + \mathbb{E}[\beta])^2 \\
&= \alpha^2\mathbb{E}[X^2] + 2\alpha\beta\mu + \beta^2 - (\alpha\mu + \beta)^2 \\
&= \alpha^2\mathbb{E}[X^2] + 2\alpha\beta\mu + \beta^2 - (\alpha^2\mu^2 + 2\alpha\mu\beta + \beta^2) \\
&= \alpha^2\mathbb{E}[X^2] + 2\alpha\beta\mu + \beta^2 - \alpha^2\mu^2 - 2\alpha\mu\beta - \beta^2 \\
&= \alpha^2\mathbb{E}[X^2] - \alpha^2\mu^2 \\
&= (\mathbb{E}[X^2] - \mu^2)\alpha^2 \\
&= (\mathbb{E}[X^2] - \mathbb{E}[X]^2)\alpha^2 \\
&= \mathrm{Var}[X]\alpha^2.
\end{aligned}
$$

Thus, we have shown that for the variance of a random variable X and a scalar α, the property $\mathrm{Var}[\alpha X + \beta] = \alpha^2\mathrm{Var}[X]$ holds.

3. We assume that we are given a random variable X with a standard normal distribution, $X \sim \mathrm{Normal}(\mu = 0, \sigma = 1)$.

   We know that $\mathbb{E}[\alpha X + \beta] = \alpha\mathbb{E}[X] + \beta$ and from part 2), $\mathrm{Var}[\alpha X + \beta] = \alpha^2\mathrm{Var}[X]$. First, let $\mathbb{E}[\bar{X}] = \mathbb{E}[\alpha X + \beta]$ and $\mathrm{Var}[\bar{X}] = \mathrm{Var}[\alpha X + \beta]$. Then, this gives us a linear system of equations, in which we need to find 2 unknown constants α, β.

   Particularly, we solve the linear system

$$
\begin{cases}
\alpha\mathbb{E}[X] + \beta = 3 \\
\alpha^2\mathrm{Var}[X] = 25
\end{cases}
$$

   So, from the first equation, we have that $\alpha\mathbb{E}[X] + \beta = 3 \implies \beta = 3$

   From the second equation, we have that $\alpha^2\mathrm{Var}[X] = 25 \implies \alpha^2 = 25 \implies \alpha = 5$.

   Hence, the transformation we need to apply to the sampled values such that they are now distributed according to $\bar{X} \sim \mathrm{Normal}(\mu = 3, \sigma = 5)$ is $\bar{X} = 5X + 3$.

4 Problem 4 [FINISHED]

We denote the probability that an islander tells the truth is $P(\text{Truth}) = \frac{1}{3}$, so the probability of a lie is $P(\text{Lie}) = \frac{2}{3}$. Furthermore, we denote the probability of an islander answering yes to our followup question, "Was that statement true?", as $P(\text{Yes})$.

First, we determine $P(\text{Yes} \mid \text{Truth})$. Given that the statement is true, then the probability of an islander saying yes is the same as the probability of the islander telling the truth, which we are given to be $\frac{1}{3}$.

Next, we determine $P(\text{Yes} \mid \text{Lie})$. Given that the statement is false, which can be considered a lie, then the probability of an islander saying yes is the same as the probability of the islander lying, which we are given to be $\frac{2}{3}$.

We will now use Bayes' Rule to calculate the probability that the statement was indeed true given that the islander answered yes to our followup question. Mathematically, we will find

$$P(\text{Truth}|\text{Yes})$$

So, using Bayes' Rule, we have that

$$\begin{aligned}
P(\mid \text{Yes}) &= \frac{P(\text{Yes} \mid \text{Truth})P(\text{Truth})}{P(\text{Yes})} \\
&= \frac{P(\text{Yes} \mid \text{Truth})P(\text{Truth})}{P(\text{Yes} \mid \text{Truth})P(\text{Truth}) + P(\text{Yes} \mid \text{Lie})P(\text{Lie})} \\
&= \frac{(\frac{1}{3})(\frac{1}{3})}{(\frac{1}{3})(\frac{1}{3}) + (\frac{2}{3})(\frac{2}{3})} \\
&= \frac{\frac{1}{9}}{\frac{5}{9}} \\
&= \frac{1}{5}.
\end{aligned}$$

Thus, we have found that $P(\text{Truth} \mid \text{Yes}) = \frac{1}{5}$.

5 Problem 5

The code for this problem can be found here.

The situation is as follows: There is one last slice of cake but mom was saving it for dessert after a long day at work. When she returns home from work, she is absolutely devastated upon finding out one of her children ate her slice of cake. It could be any one of her children, and she has no idea which it could be, thus indicating a uniform distribution. Additionally, in terms of remaining clean utensils, there was only one spoon, one fork, and one set of chopsticks left, as mom planned to load the dishwasher after coming home from work. Thus, the possible utensils is also uniform distribution.

OUTPUT

```
num_sims = 10:
annie_chopsticks occurred with a probability of 0.2.
annie_spoon occurred with a probability of 0.1.
gadzi_chopsticks occurred with a probability of 0.1.
gadzi_spoon occurred with a probability of 0.2.
ren_chopsticks occurred with a probability of 0.1.
ren_fork occurred with a probability of 0.1.
ren_spoon occurred with a probability of 0.2.


num_sims = 100:
annie_chopsticks occurred with a probability of 0.12.
annie_fork occurred with a probability of 0.08.
annie_spoon occurred with a probability of 0.18.
gadzi_chopsticks occurred with a probability of 0.1.
gadzi_fork occurred with a probability of 0.1.
gadzi_spoon occurred with a probability of 0.14.
ren_chopsticks occurred with a probability of 0.11.
ren_fork occurred with a probability of 0.11.
ren_spoon occurred with a probability of 0.06.


num_sims = 1000:
annie_chopsticks occurred with a probability of 0.12.
annie_fork occurred with a probability of 0.11.
annie_spoon occurred with a probability of 0.1.
gadzi_chopsticks occurred with a probability of 0.11.
gadzi_fork occurred with a probability of 0.12.
gadzi_spoon occurred with a probability of 0.12.
ren_chopsticks occurred with a probability of 0.11.
ren_fork occurred with a probability of 0.11.
ren_spoon occurred with a probability of 0.1.


num_sims = 10000:
annie_chopsticks occurred with a probability of 0.11.
annie_fork occurred with a probability of 0.1.
annie_spoon occurred with a probability of 0.11.
gadzi_chopsticks occurred with a probability of 0.12.
gadzi_fork occurred with a probability of 0.11.
gadzi_spoon occurred with a probability of 0.11.
ren_chopsticks occurred with a probability of 0.11.
ren_fork occurred with a probability of 0.11.
ren_spoon occurred with a probability of 0.11.


num_sims = 100000:
annie_chopsticks occurred with a probability of 0.11.
annie_fork occurred with a probability of 0.11.
annie_spoon occurred with a probability of 0.11.
gadzi_chopsticks occurred with a probability of 0.11.
gadzi_fork occurred with a probability of 0.11.
gadzi_spoon occurred with a probability of 0.11.
ren_chopsticks occurred with a probability of 0.11.
```

From the output, we observe that $N = 10000$ is when the approximate probabilities to become almost indistinguishable from the analytic ones.

Now, we will justify this with the following tables from our code:

Simulated joint table:

```
Simulated joint table (N=10000):
         spoon    fork  chopsticks
gadzi   0.1174  0.1087      0.1154
annie   0.1107  0.1110      0.1083
ren     0.1101  0.1093      0.1091
```

Analytic joint table:

```
Analytic joint table:
           spoon      fork  chopsticks
gadzi   0.111111  0.111111    0.111111
annie   0.111111  0.111111    0.111111
ren     0.111111  0.111111    0.111111
```

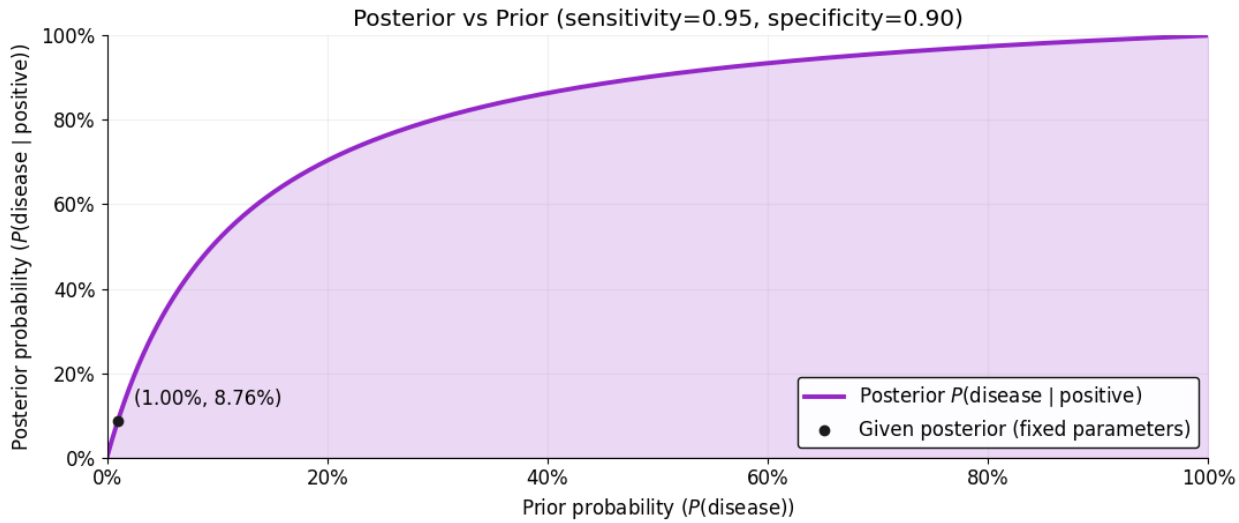
Absolute difference in tables:

```
Absolute error |sim - analytic|:
           spoon      fork  chopsticks
gadzi   0.006289  0.002411    0.004289
annie   0.000411  0.000111    0.002811
ren     0.001011  0.001811    0.002011
Max absolute error: 0.006288888888888899
```

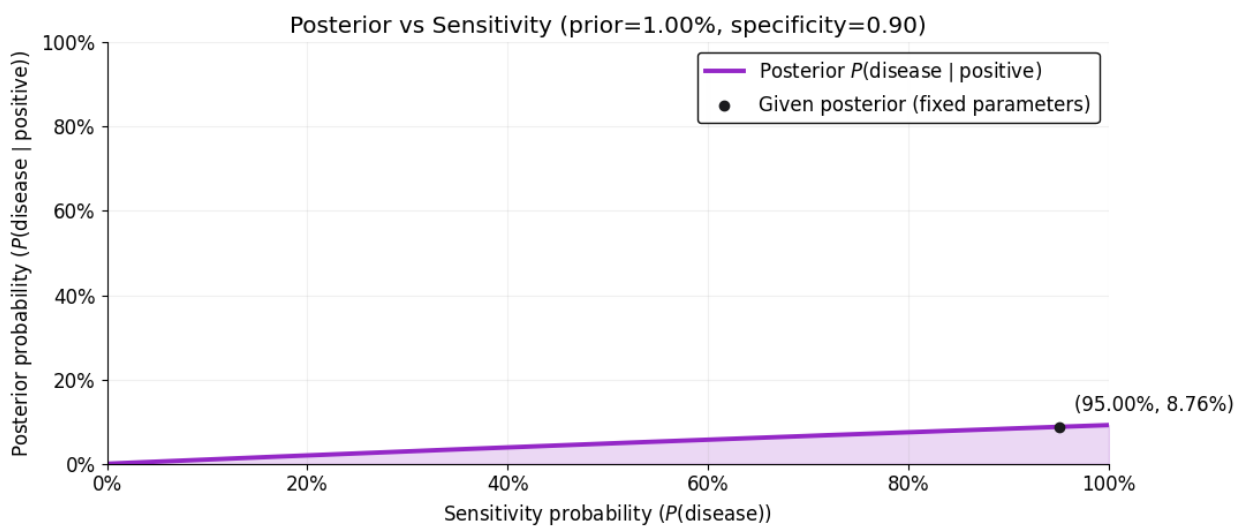
6 Problem 6 [FINISHED]

The code for this problem can be found here.

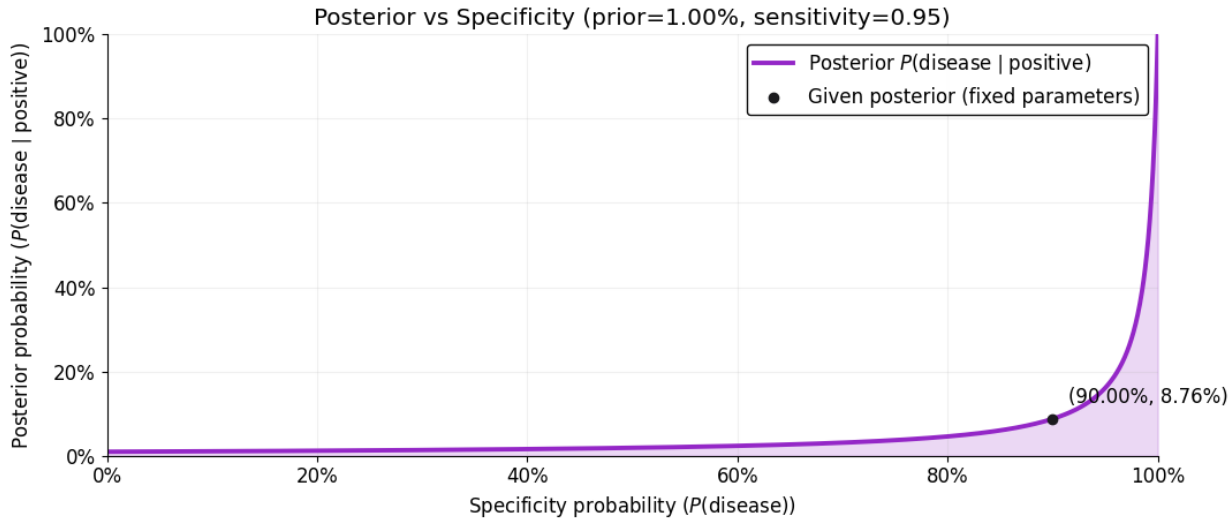
1. OUTPUT



2. OUTPUT



3. OUTPUT



Analysis:

From the above graphs, we observe that they all exhibit monotonic increasing functions.

As a function of prior, the posterior rapidly increases and then flattens out. This illustrates how as a disease becomes common, a positive test result becomes much more informative. With small values of prior probability, the posterior is dominated by false positives.

As a function of sensitivity, we observe slower growth. We observe that increasing sensitivity probability increases posterior, due to the fact that true positives become more and more likely. When the disease is rare, there are not much improvements to be made by sensitivity.

As a function of specificity, we notice a similar trend as a function of sensitivity. However, the main difference is that even with very low small specificity probability, the posterior is not as low, starting at around 20%. With higher specificity probability, it reduces false positives which allows for the posterior to be instead dominated by true positives.

7 Problem 7 [FINISHED]

The code for this problem can be found here.

OUTPUT

```
Case: spherical
  max |pdf diff|   = 0.0
  max |log diff|   = 0.0
  mean |log diff|  = 0.0

Case: diagonal
  max |pdf diff|   = 4.336808689942018e-18
  max |log diff|   = 3.552713678800501e-15
  mean |log diff|  = 6.217248937900876e-16

Case: full
  max |pdf diff|   = 6.938893903907228e-18
  max |log diff|   = 2.842170943040401e-14
  mean |log diff|  = 1.3233858453531866e-14

Class log_pdf vs SciPy logpdf, max diff: 7.460698725481052e-14
Sample mean error (norm): 0.04315871682402934
Sample cov error (Fro):   0.11092280367557682
```

We verified the implementation by comparing the outputs of the LLM's code with that of scipy. Particularly, it is the output of pdf and logpdf. From the image of the output, we observe that the LLM performed quite well, since the difference between the LLM's implementation and Scipy's implementation are on the magnitude of no more than 10^{-14}, indicating very small marginal errors, possibly floating point precision errors.

Additionally, we verified the sampling method by generating random samples and confirming that the empirical mean and covariance converge to the specified parameters. We observe that the deviations were consistent with that of the Monte Carlo sampling error.